Jiashen Ren

Ph.D. Student in Data Science, City University of Hong Kong

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Education

City University of Hong Kong Aug. 2026 – Present
Ph.D. in Data Science
Advisor: Prof. Xiangyu Zhao; Applied Machine Learning (AML) Lab

Zhejiang University Sep. 2022 – May 2026
B.S. in Electrical Engineering

University of Illinois at Urbana-Champaign Sep. 2022 – May 2026
B.S. in Electrical Engineering

Research Interests

Human-centered and trustworthy AI; LLM behavior, evaluation, and security; constraint-aware machine learning.

Publications

[1] Haoyu Zhu, Yao Zhang, **Jiashen Ren**, and Qingchun Hou. **T-SKM-Net: Trainable Neural Network Framework for Linear Constraint Satisfaction via Sampling Kaczmarz-Motzkin Method.** *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2026.

Research in Progress

Construct-Surgicality of Persona Prompts in LLM User Simulation

- Studying whether persona prompts activate intended traits without shifting construct-distal preferences in LLM-based user simulation.
- Using trait-by-domain audits, null controls, and factorization tests to distinguish target-trait activation from cross-domain leakage across API-served and open-weight models.
- Examining how structured discriminant evidence and task design attenuate leakage while retaining target traits, with downstream evaluation in synthetic surveys and A/B choices.

Research Experience

Constraint-Oriented Neural Networks for Power-System Optimization
SRTP Research Project Feb. 2025 – Aug. 2026
Supervisor: Prof. Qingchun Hou, ZJU-UIUC Institute, Zhejiang University

- Designed and implemented a physics-constrained neural network architecture for linear constraint satisfaction in power-system optimization.
- Worked on graph neural network-based grid decomposition and boundary-flow prediction, with validation on IEEE test systems for AC optimal power flow.

Research Experience (continued)

DiffSinger-Based Voice Enhancement and Synthesis
Summer Research Jun. 2023 – Aug. 2023
Supervisor: Prof. Zhou Zhao, College of Computer Science, Zhejiang University

- Reproduced DiffSinger components, including noise scheduling and denoising networks, and studied the noise-to-Mel-spectrogram generation pipeline.
- Assisted with experimental analysis and manual annotation and evaluation of synthesized audio quality.

Selected Technical Project

FPGA-Based Handwritten Digit Recognition Code
Course project (ECE385)

- Implemented a 1,024–32–10 multilayer perceptron in SystemVerilog, converting a PyTorch model to Q7.24 fixed-point arithmetic with custom multiply-accumulate units.
- Integrated camera input, image processing, VGA display, and I^2C communication; achieved timing closure at 50 MHz.

Teaching Experience

Calculus III (MATH241) Sep. 2025 – Dec. 2025
Teaching Assistant

Introduction to Differential Equations (MATH285) Jan. 2025 – May 2025
Teaching Assistant

Selected Honors and Awards

Honorable Mention, Mathematical Contest in Modeling (MCM) 2024

National Silver Medal, China International College Students' Innovation Competition 2023

Technical Skills

Programming: Python, C, SystemVerilog.

Machine learning: PyTorch, PyTorch Lightning, NumPy, pandas, scikit-learn.

Tools: Git, Linux, LaTeX, Intel Quartus.

Languages: Mandarin (native); English (proficient).